Harrison and Kreps (1978)

$\Omega(s)$ expected return under true probabilities of holding equity in state s and selling when the state switches to s', $s' \neq s$

$\pi(s,s)$ true Markov probability of remaining in state s

$$\Omega(s) = \frac{\sum_{t=1}^{\infty} \beta^{t-1} \pi(s,s)^{t-1} \left\{ \pi(s,s)\, d(s) + [1-\pi(s,s)][d(s') + \bar{p}(s')] \right\}}{\bar{p}(s)}$$

$$= \frac{\pi(s,s)\, d(s) + [1-\pi(s,s)][d(s') + \bar{p}(s')]}{[1-\beta\pi(s,s)]\, \bar{p}(s)}$$

$$\Omega(1) = \frac{[1-\pi(1,1)][1+\bar{p}(2)]}{[1-\beta\pi(1,1)]\, \bar{p}(1)}$$

$$\Omega(2) = \frac{\pi(2,2) \cdot 1 + [1-\pi(2,2)]\, \bar{p}(1)}{[1-\beta\pi(2,2)]\, \bar{p}(2)}$$

$$\Omega(1)\Big|_{\pi(1,1)=\pi_a(1,1)} = \Omega(2)\Big|_{\pi(2,2)=\pi_b(2,2)} = \frac{1}{\beta}$$